

1. (15%)
 - A. (8%) Find the rectangle of minimum perimeter with area a^2 . That is, minimize $L(x, y) = 2x + 2y$ subject to $f(x, y) = xy - a^2 = 0$.
 - B. (7%) Sketch the contours of $L(x, y)$ and the constraint.
2. (15%) Consider the bilinear discrete-time system $x_{k+1} = Ax_k + Dx_k u_k + bu_k$, where $x_k \in \mathbb{R}^n$, $u_k \in \mathbb{R}$, with quadratic performance index $J = \frac{1}{2} x_N^T S_N x_N + \frac{1}{2} \sum_{k=0}^{N-1} (x_k^T Q x_k + r u_k^2)$, where $S_N \geq 0$, $Q \geq 0$, $r > 0$.
 - A. (5%) Find the optimal control is the bilinear state-costate feedback, u_k .
 - B. (10%) Find the state and costate equations after eliminating u_k .
3. (20%) The scalar plant $\frac{d}{dt}x = u$ has performance index $J(t_0) = \frac{1}{2} s x^2(t_0) + \frac{1}{2} \int_{t_0}^T r u^2 dt$.
 - A. (10%) Find a closed-loop control minimizing J .
 - B. (10%) Find an open-loop control that minimizes J and drives the state to a given $x(T)$. Initial state $x(t_0)$ is known.
4. (15%) An aircraft can fly from left to right along the paths shown in Fig. 1. Intersections a, b, c, . . . represent cities, and the numbers represent the fuel required to complete each path. Let us use the principle of optimality to find the maximum fuel route from a to i.

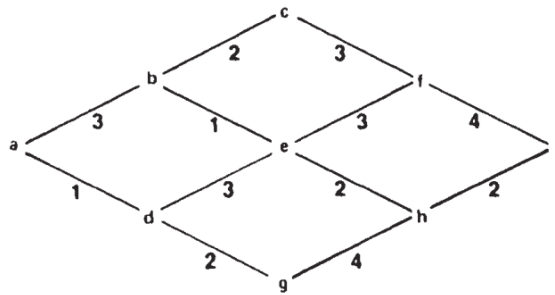


Fig. 1

5. (35%) True or False. If true, please state the main reason and prove it. Otherwise, please correct the statements or provide a counter example.
 - A. (5%) The dynamic programming approach for finding optimal policies is usually achieved by using causal experiences received stagewise forward in time.
 - B. (5%) Both the Policy Iteration (PI) algorithm and the Value Iteration (VI) algorithm for applying general deterministic dynamical systems can be implemented online in real time by observing data measured along the system trajectories.
 - C. (5%) PI converges to the optimal value in fewer steps, since it does more work in updating the value at each step. On the other hand, VI is the easiest to implement as it takes only one iteration in value updating.
 - D. (5%) The main advantage of the Q-function approach is that minimizing the Bellman optimality equation requires knowledge only of the Q-function and not the system dynamics.
 - E. (5%) Unlike Monte Carlo Simulations, Temporal difference (TD) RL uses only one sample path, namely the current system trajectory, to update the value. Then, TD methods will update the value estimate to make the TD error small.
 - F. (10%) For DT linear quadratic regulator (DT LQR) problems under TD RL, show that the Bellman equation (11.5-7) in the textbook is a linear Lyapunov equation and that (11.5-5) is the DT algebraic Riccati equation (ARE).